

The Use of the Integral Method to Compare the Efficiency of Image Processing Algorithms Depending on Size and Format Using the Example of Road Photos

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Abstract— Roads occupy one of the most important places in the life of a modern person. The quality of the road surface is its key characteristics. There are many different systems for assessing the quality of roads, that mainly solve the problem using image analysis. One of the key stages of such systems is image processing. Processing, for example, includes edge detection and segmentation, that can be implemented using both machine learning algorithms and traditional image processing methods. In this paper, a comparative analysis of the traditional image processing methods, depending on size and format, using the integral method is carried out. Edge detection and segmentation are categories that are selected for analysis. 6 edge detection methods (Canny edge detection, Kirsch edge detection, Laplace operator, Marr-Hildreth edge detection, Prewitt operator, Sobel operator) and 4 segmentation algorithms (Adaptive thresholding, Global thresholding, K-means clustering, Linear clustering) are considered. The research material consists of road surface images in 5 sizes (447x447, 632x632, 775x775, 894x894 and 1000x1000) and 3 formats (bmp, jpg, png). The efficiencies of the analyzed methods are determined using an integral approach combining time and space indicators. The best integral efficiencies results demonstrate among the edge detection algorithms Prewitt operator and Sobel operator; Global thresholding is the most effective segmentation method.

Keywords—*image processing, algorithm, efficiency, integral method, size, format, road photo*

I. INTRODUCTION

Roads occupy one of the most important places in the life of a modern person. The quality of the road surface is its key characteristics. There are many different systems for assessing the quality of roads, which mainly solve the problem using image analysis. One of the key stages of such systems is image processing. Processing, for example, includes edge detection and segmentation, which can be implemented using both traditional image processing methods [1-3] and machine learning algorithms [4, 5].

Currently, the most popular approach to image processing is the use of neural networks. However, the requirements for image processing methods are consistently changing: new quality assessment criteria are being added. Therefore, to complete the picture, it is necessary to compare as many methods as possible, including traditional image processing. In this paper, the efficiency analysis is devoted to traditional

algorithms. In subsequent studies, machine learning methods will already be evaluated using the integral approach.

This article analyzes the existing traditional image processing algorithms, using the example of edge detection and segmentation methods. The main characteristics that are considered when determining the efficiency of the algorithm are the average image processing time (time indicator) and the average allocated memory (space indicator). The most effective image processing method is one that successfully combines both. An approach that considers both time and space indicators is called the integral method.

The **purpose** of the work is to analyze the efficiency of image processing algorithms using the integral method.

The **research material** is road surface photos in various sizes and formats.

The **scientific novelty** lies in the integral approach for evaluating the efficiency of image processing methods.

The research objectives:

1. To provide brief information about the selected image processing algorithms.
2. To describe the methodology of conducting an experiment to obtain the efficiencies of the analyzed algorithms based on an integral method.
3. To analyze the results obtained and draw the appropriate conclusions from the study.

II. IMAGE PROCESSING ALGORITHMS

Image processing algorithms of 2 categories are selected for the analysis: edge detection and segmentation.

A. Edge detection

Edge detection is an important task in image processing and computer vision. It involves identifying and localizing sharp breaks in an image, which usually correspond to significant changes in intensity or color [6].

The following 6 methods are used for comparison: *Canny edge detection (CED, Canny)* [7], *Kirsch edge detection (KED, Kirsch)* [8], *Laplace operator (Laplace)* [9], *Marr-Hildreth edge detection (MHED, Marr-Hildreth)* [10], *Prewitt operator (Prewitt)* [11], *Sobel operator (Sobel)* [12].

B. Segmentation

Image segmentation is one of the key tasks of computer vision. It allows you to separate objects, borders, or structures within an image for more meaningful analysis [13].

The following 4 algorithms are selected for analysis: *Adaptive thresholding* [14], *Global thresholding* [15], *K-means clustering* [16], *Linear clustering* [17].

III. METHODOLOGY

For the experiment, 10 road surface images for 5 different sizes (447x447, 632x632, 775x775, 894x894 and 1000x1000) in bmp, jpg and png formats are used. The photo sizes are selected in such a way that the number of pixels in the image varies evenly. The listed sizes in pixels: 200000 ($\approx 447 \times 447$), 400000 ($\approx 632 \times 632$), 600000 ($\approx 775 \times 775$), 800000 ($\approx 894 \times 894$), 1000000 (1000x1000). The change step is 200000 pixels.

The experimental computer has following characteristics: operating system – Windows 10 Pro, system type – x64, RAM – 8 GB, hard drive type – basic, RAM channel – single, RAM type – DDR4.

C# is used as the programming language for the experiment, Visual Studio 2022 has been selected as the development environment. The library "System.Diagnostics" is used to record time and memory allocation.

The experiment that determines the efficiency of the selected image processing algorithms has the following stages:

1. The photos described above are used as research material for each of the analyzed algorithms (6 edge detection and 4 segmentation methods).
2. For 10 selected methods, 10 different images are improved for each of the 3 formats and 5 resolutions. 10 photos for each experiment are stored in a separate computer folder (150 different folders in total: 10 (selected algorithms) x 3 (formats) x 5 (resolutions)). The processed versions of the images are placed in a new folder located in the same location as the experimental one.
3. For each experiment (processing 10 different photos), the total execution time (in seconds) and the total allocated memory (in megabytes) are recorded using "System.Diagnostics" library. To find their average values, each of the listed indicators is divided by the number of images in the folder (10 in total). The formulas are presented below.

$$t_{avr} = t_{total}/N \quad (1)$$

where t_{avr} is average time to obtain a processed image version, t_{total} is total time spent for one experiment, N is number of experimental photos (equal to 10).

$$s_{avr} = s_{total}/N \quad (2)$$

where s_{avr} is the average memory used to obtain a processed version, s_{total} is the total memory used during one experiment.

IV. RESULTS AND DISCUSSION

The average time and allocated memory on obtaining a processed image in various formats and resolutions for edge detection and segmentation algorithms are shown in Tables I and II. The values are rounded to 3 decimal places.

TABLE I. AVERAGE TIME (IN SECONDS) TO OBTAIN A PROCESSED IMAGE IN VARIOUS FORMATS AND RESOLUTIONS FOR SELECTED IMAGE PROCESSING ALGORITHMS

BMP					
Algorithm	Pixels				
	200000	400000	600000	800000	1000000
Canny	0.788	2.144	4.828	7.567	11.727
Kirsch	0.256	0.425	0.684	0.854	1.039
Laplace	0.095	0.134	0.193	0.253	0.313
Marr-Hildreth	0.465	1.661	4.192	6.331	10.192
Prewitt	0.047	0.107	0.144	0.205	0.243
Sobel	0.051	0.104	0.226	0.193	0.313
Adaptive thresholding	2.349	5.163	8.020	10.950	13.616
Global thresholding	0.015	0.024	0.037	0.058	0.057
K-means clustering	0.301	0.614	0.974	1.113	1.580
Linear clustering	1.434	4.275	8.185	13.396	18.330
JPG					
Algorithm	Pixels				
	200000	400000	600000	800000	1000000
Canny	0.626	1.984	4.608	7.312	11.301
Kirsch	0.258	0.429	0.786	0.944	1.122
Laplace	0.129	0.142	0.276	0.351	0.480
Marr-Hildreth	0.554	1.642	4.022	6.360	9.908
Prewitt	0.048	0.107	0.142	0.198	0.320
Sobel	0.058	0.108	0.156	0.276	0.248
Adaptive thresholding	2.660	5.429	8.148	11.080	14.195
Global thresholding	0.014	0.024	0.038	0.055	0.070
K-means clustering	0.380	0.879	0.700	1.108	2.017
Linear clustering	1.311	4.198	8.368	13.119	17.920
PNG					
Algorithm	Pixels				
	200000	400000	600000	800000	1000000
Canny	0.731	2.020	4.925	7.372	11.094
Kirsch	0.170	0.509	0.676	0.948	1.198
Laplace	0.065	0.219	0.278	0.330	0.440
Marr-Hildreth	0.545	1.580	4.075	6.241	9.865
Prewitt	0.048	0.108	0.140	0.204	0.311
Sobel	0.048	0.175	0.204	0.207	0.326
Adaptive thresholding	2.792	5.628	8.400	10.548	13.611
Global thresholding	0.023	0.028	0.035	0.045	0.059
K-means clustering	0.276	0.563	0.707	1.172	1.373
Linear clustering	1.309	4.093	7.883	13.003	17.673

TABLE II. AVERAGE ALLOCATED MEMORY (IN MEGABYTES) WHEN OBTAINING A PROCESSED IMAGE IN VARIOUS FORMATS AND RESOLUTIONS FOR SELECTED IMAGE PROCESSING ALGORITHMS

BMP					
Algorithm	Pixels				
	200000	400000	600000	800000	1000000
Canny	76.738	139.922	148.082	182.367	272.566
Kirsch	46.746	56.082	64.672	72.621	81.246
Laplace	50.762	51.375	56.910	56.91	62.125
Marr-Hildreth	87.418	80.750	102.562	216.641	143.844
Prewitt	79.215	48.559	52.234	56.043	60.031
Sobel	48.414	49.160	52.082	55.809	60.039
Adaptive thresholding	43.508	46.320	50.027	54.898	56.711
Global thresholding	445.996	424.773	360.715	302.316	203.402
K-means clustering	48.621	52.18	61.953	68.605	73.289
Linear clustering	54.246	62.848	72.922	97.051	106.359
JPG					
Algorithm	Pixels				
	200000	400000	600000	800000	1000000
Canny	254.785	136.344	143.57	177.48	262.332
Kirsch	55.414	54.613	62.651	69.312	76.031
Laplace	53.645	49.500	52.988	54.586	57.543
Marr-Hildreth	100.016	77.965	100.148	210.484	139.324
Prewitt	50.789	46.832	49.316	52.375	55.449
Sobel	50.797	46.570	49.051	52.094	55.172
Adaptive thresholding	49.746	46.328	48.133	49.926	52.551
Global thresholding	53.617	52.035	56.344	61.066	65.215
K-means clustering	54.250	53.449	55.758	61.008	73.156
Linear clustering	80.258	59.25	68.496	88.824	96.457
PNG					
Algorithm	Pixels				
	200000	400000	600000	800000	1000000
Canny	476.066	258.754	145.547	195.225	264.812
Kirsch	51.426	56.926	63.664	70.859	78.551
Laplace	50.031	51.660	55.395	56.684	59.930
Marr-Hildreth	96.363	79.359	101.164	212.727	139.867
Prewitt	47.184	48.988	50.760	54.102	57.594
Sobel	47.188	49.000	51.004	54.410	57.375
Adaptive thresholding	46.488	48.012	49.387	51.379	55.137
Global thresholding	49.422	52.781	57.227	65.316	67.441
K-means clustering	48.465	51.777	61.895	64.820	74.117
Linear clustering	76.809	60.480	70.043	91.086	99.906

To make the results presented in Tables I and II more visual, corresponding graphics of the dependence of average

time and allocated memory on resolution and format have been plotted. The graphics are demonstrated in Fig. 1 and 2. They were plotted using the “pyplot” library of the Python programming language.

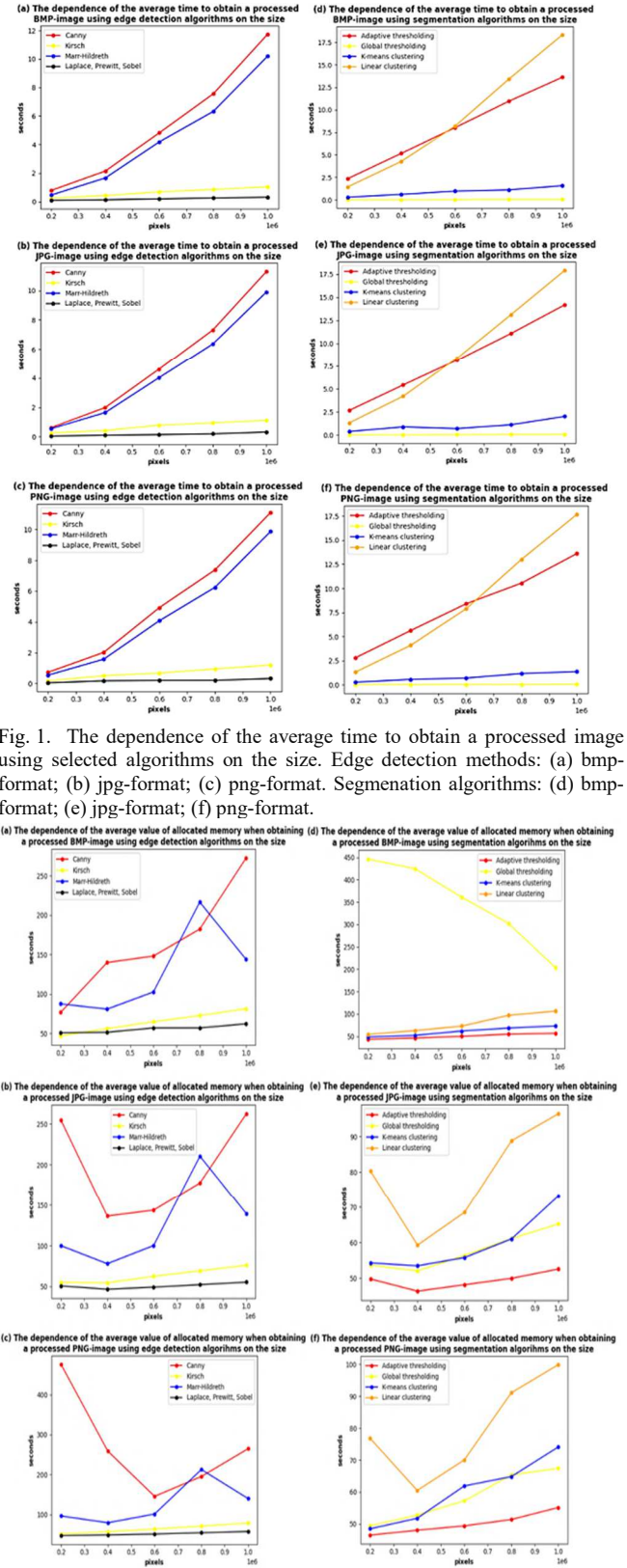


Fig. 1. The dependence of the average time to obtain a processed image using selected algorithms on the size. Edge detection methods: (a) bmp-format; (b) jpg-format; (c) png-format. Segmentation algorithms: (d) bmp-format; (e) jpg-format; (f) png-format.

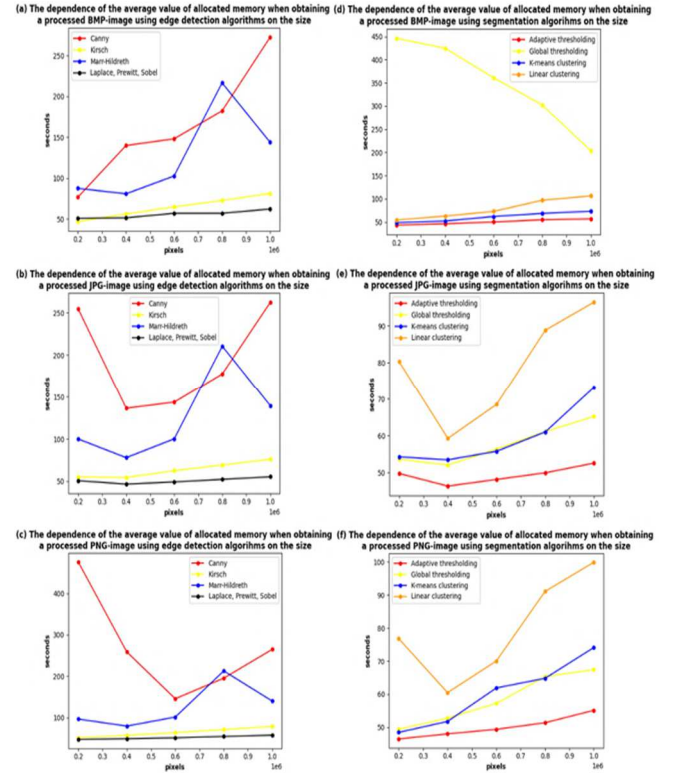


Fig. 2. The dependence of the average value of allocated memory when obtaining a processed image using selected algorithms on the size. Edge detection methods: (a) bmp-format; (b) jpg-format; (c) png-format. Segmentation algorithms: (d) bmp-format; (e) jpg-format; (f) png-format.

Based on the graphics, the following conclusions can be drawn:

- Fig. 1. Kirsch edge detection, Laplace operator, Prewitt operator and Sobel operator have a linear $O(n)$ dependence of the average time to obtain an improved image version on the photo size; Canny edge detection and Marr-Hildreth edge detection have a quadratic dependence $O(n^2)$ [18]. Adaptive thresholding, Global thresholding and K-means clustering have a linear $O(n)$ dependence for both the average time to obtain a processed image on the photo size; Linear clustering have quadratic dependence $O(n^2)$.
- Fig. 2. Kirsch edge detection, Laplace operator, Prewitt operator and Sobel operator have a linear $O(n)$ dependence of the average allocated memory when obtaining a processed image on the photo size; Canny edge detection and Marr-Hildreth edge detection and all selected segmentation algorithms don't have a particular dependence.

The time/space efficiency is calculated by summing the ratio of the current time/space indicator to the minimum value among all image processing algorithms in the category (edge detection or segmentation) for the current photo size and format, followed by dividing the product of the sizes (5) and formats (3) counts (total number of terms). The formulas 3 and 4 are presented below.

$$E_{T,i} = (\sum_{k=1}^F \sum_{j=1}^P t_{i,j,k \text{ avr}} / t_{j,k \text{ avr min}}) / (P \cdot F) \quad (3)$$

$$E_{S,i} = (\sum_{k=1}^F \sum_{j=1}^P S_{i,j,k \text{ avr}} / S_{j,k \text{ avr min}}) / (P \cdot F) \quad (4)$$

where $E_{T,i}$, $E_{S,i}$ is the time/space efficiency of the algorithm i , $t_{i,j,k \text{ avr}}$, $S_{i,j,k \text{ avr}}$ is the average (processing time)/(allocated memory) of an image of size j and format k using the method i , $t_{j,k \text{ avr min}}$ is the minimum (processing time)/(allocated memory) of a photo of size j and format k among all method, P is the sizes count (5), F is the formats count (3).

Let's assume that time and space indicators have the same impact. Then each component has the same weight, since there are 2 of them, the value is 1/2. Then the integral efficiency has the following formula (5):

$$E_{Int,i} = (E_{T,i} + E_{S,i}) / 2 \quad (5)$$

where $E_{Int,i}$ is the integral efficiency of algorithm i .

All efficiencies (time, space and integral) of the selected image processing algorithms are presented in Tables III. The values are rounded to 4 decimal places for time and space efficiencies, and to 5 digits for integral efficiencies.

TABLE III. THE EFFICIENCIES (TIME, SPACE AND INTEGRAL) OF THE IMAGE PROCESSING ALGORITHMS

Algorithm	Efficiency		
	Time	Space	Integral
Canny	0.0399	0.2964	0.16815
Kirsch	0.2215	0.8397	0.53060
Laplace	0.6225	0.9449	0.78370
Marr-Hildreth	0.0500	0.4554	0.25270
Prewitt	0.9792	0.9705	0.97485
Sobel	0.8683	0.9962	0.93225
Adaptive thresholding	0.0049	1.0000	0.50245
Global thresholding	1.0000	0.6279	0.81395
K-means clustering	0.0437	0.8379	0.44080
Linear clustering	0.0060	0.6504	0.32820

The Prewitt operator and Sobel operator have the best both time and capacity indicators. Therefore, these algorithms have the best integral efficiencies. Adaptive thresholding has the

best space efficiency, Global thresholding has the best both time and integral efficiencies.

For a clearer representation of the results obtained in Table III, the efficiencies diagrams for all image processing algorithms and their average values are plotted. Diagrams are presented in Fig. 3.

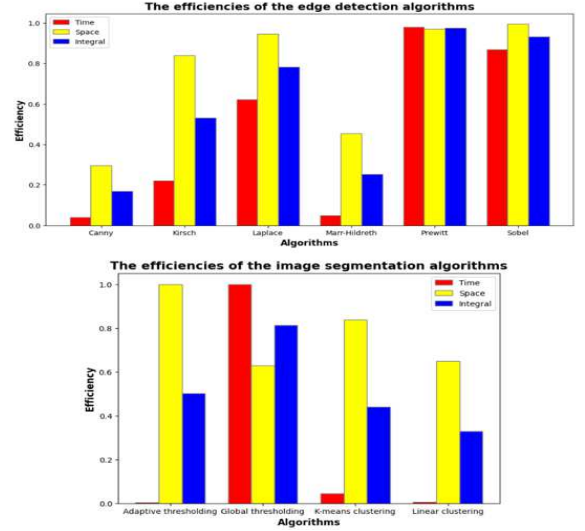


Fig. 3. The efficiencies diagrams of the edge detection algorithms.

A similar study on the analysis of traditional image processing methods was conducted in [1]. It considered some of the methods analyzed in this work: Kirsch, Laplace, Prewitt. The MSE (Mean squared error) and PSNR (Peak signal-to-noise ratio) indicators were also compared. Our work examines more image processing algorithms (10 in total), as well as analyzing other indicators: time and space characteristics.

It was not possible to find studies devoted to evaluating the efficiencies of neural networks (for example, CNN), in which time or space are used as the main characteristics (mainly, such indicators as precision and recall are evaluated [4]). Therefore, such an analysis will most likely have to be carried out independently in the future.

V. CONCLUSION

Evaluating the results obtained in the course of the work, the following conclusions can be drawn:

- For a more complete study, there is not enough comparison with machine learning algorithms. However, an analysis of their efficiencies is planned to be carried out in the following works.
- New performance indicators, such as fault tolerance, can be added to the integral method. The weights for a specific indicator can also be changed (for example, the time indicator is 0.6, the space indicator is 0.4).
- For more accurate measures, the number of experimental images can be increased. For example, from 10 to 100.

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